- In response to the 1973 oil crisis, the Philippines began construction of a nuclear power plant. The 621 MWe Westinghouse unit at Bataan was completed in 1984 but never commissioned.
- With electricity demand expected to more than triple by 2040, energy security is a stated priority for the government.
- The country aims to have its first nuclear power plant operational by 2032. It is exploring the possible revival of

Privacy - Terms

the Bataan project or constructing a small modular reactor.

Electricity sector

Total generation (in 2023): 118 TWh

Generation mix: coal 73.8 TWh (62%); natural gas 16.7 TWh (14%); geothermal 10.7 TWh (9%); hydro 10.3 TWh (9%); solar 2.6 TWh (2%); biofuels & waste 1.4 TWh; oil 1.3 TWh; wind 1.3 TWh.

Import/export balance: no imports or exports

Total consumption: 95.8 TWh

Per capita consumption: c. 800 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

The Philippines' electricity demand is growing rapidly. Per capita electricity consumption in the country has increased from about 600 kWh in 2000.

The country generates about 80% of its electricity from fossil fuels. Its population of about 115 million was served by power generation capacity of only 28.3 GWe in 2023. The Philippine Department of Energy (DOE) expects peak electricity demand to more than triple by 2040.

A key priority for the government is energy security, as emphasized in the Philippine Energy Plan (PEP). Self-sufficiency is becoming more challenging as demand for oil from the transport sector increases, and demand for coal for power generation grows as the country's main gas field, Malampaya, is progressively depleted.

PEP 2017-2040 set out a target for universal access to electricity by 2022, but as of June 2023, 91.1% of households had access to electricity.

Nuclear power industry

Nuclear energy plans

In 2007 the Philippines DOE set up a project to study the development of nuclear energy, in the context of an overall energy plan for the country. Nuclear energy would be considered in order to reduce the country's dependency on imported oil and coal. In its 2008 update of the national energy plan, 600 MWe nuclear was projected online in 2025, with further 600 MWe increments in 2027, 2030 and 2034 to give 2400 MWe.

Apart from Bataan, the government was considering two further 1000 MWe Korean Standard Nuclear Plant units, using equipment from the aborted North Korean KEDO project. Kepco was reported to be offering this equipment for $1.1 billion, but it has since been redeployed elsewhere.

The government established a working group with a view to proceeding along the same lines as Thailand, retaining engineering consultants to guide progress. At the end of 2012 the DOE confirmed that nuclear remained a live option of the grounds of costs and clean air.

Early in 2016, and again in November, the DOE reiterated that nuclear power was a live option, possibly to take over some of coal's base-load role.

In October 2016 the DOE created a nuclear energy programme implementing organization (NEPIO) to assess the feasibility of nuclear power as a long-term energy option in accordance with the IAEA 'milestones' approach. In the Philippine Energy Plan (PEP) 2018-2040, the DOE presented its Nuclear Power Program Roadmap, which envisages the country's first nuclear plant being commissioned from 2027 at the earliest. The first of the roadmap's three phases consisted of a number of pre-feasibility studies to identify critical infrastructure areas to be addressed before launching a nuclear power programme.

In March 2017 the DOE said it was exploring the potential of a small reactor in Sulu province, Mindanao, and would produce an overall nuclear programme for the country, including Bataan. In May 2017 a nuclear cooperation agreement was signed with Rosatom, followed by another in November 2017 setting out a five-year programme of discussions. The DOE signed a further agreement with Rosatom in October 2019, to assess the feasibility of a small nuclear power plant, floating or on land, and probably using RITM-200 reactors.

In June 2018 the DOE announced that KHNP would carry out a study on building a 100 MWe SMART small modular reactor (SMR) in Cagayan Economic Zone Authority (CEZA) in Santa Ana, Cagayan, on the northeasternmost point of Luzon, as well as a review of the Bataan plant. In October 2019 the DOE signed a memorandum of intent with Rusatom Overseas to carry out a pre-feasibility study on building SMRs in the Philippines.

In July 2020 the President signed Executive Order 116, which established the Nuclear Energy Program Inter-Agency Committee (NEP-IAC) to conduct a study on a national position on a nuclear energy programme. The President stated: "There is an imperative need to revisit the country's policy on nuclear energy and to determine its feasibility as a long-term option for power generation."

In July 2021 the DOE announced that it had identified 15 possible locations that could host a nuclear power plant.

In February 2022 the President signed Executive Order 164, which mandates the DOE to develop and implement a nuclear programme. The order requires the NEP-IAC to "make recommendations on the use and viability of the Bataan nuclear

power plant and the establishment of other facilities for the utilization of nuclear energy."

In November 2022 DOE Undersecretary of Energy Sharon Garin said that it would take at least a decade to build a new nuclear power plant.

In January 2023 preparations for a 123 Agreement – establishing legal framework for the transfer of US-origin special nuclear material, as well as the export of nuclear fuel, reactors, and equipment for peaceful use – were announced between the U.S. and Philippines.

In February 2024 the DOE established the Nuclear Energy Program Coordinating Committee implement the country's nuclear energy programme. The new committee is tasked with drafting recommendations for upgrades of the Philippines transmission system to accommodate new nuclear capacity on the grid, nuclear-supportive national legislation, siting requirements for possible nuclear power plant locations, and environmental protection.

In October 2025 the DOE released a circular that seeks to establish the conditions for investment in the country's first nuclear capacity. The DOE said that the first commercially developed and operated nuclear power plant, designated as the Pioneer NPP, will be treated as a baseload facility and granted priority dispatch in coordination with the DOE, the Independent Market Operator, and the System Operator.

To further support long-term financial viability, flexible contracting mechanisms such as auctions, direct contracting, or aggregation for industrial and economic zone use will be introduced. In parallel, the Energy Regulatory Commission, in consultation with stakeholders, will implement a Regulatory Asset Base (RAB)-type model or a similar capital recovery mechanism, anchored on minimum contract terms of 25 years, extendable for another 25 years.

Bataan nuclear power plant

In response to the 1973 oil crisis, the Ferdinand Marcos administration decided to build a nuclear plant comprising two 600 MWe units. By the time the contract was signed in February 1976, it was for a single 621 MWe Westinghouse unit at Napot Point in Bataan. Construction work began in July 1976 and was completed in 1984 at a cost of $1.9 billion. However, due to financial issues and safety concerns related to earthquakes, the plant was never

loaded with fuel or operated. Following the April 1986 Chernobyl accident, recently-elected President Corazon Aquino decided to mothball the plant.

The Bataan project was beset by allegations of corruption. In 1988 the Philippines government brought charges of bribery against Westinghouse and engineering firm Burns & Roe Associates; however, investigations by the US Justice Department and the Securities and Exchange Commission found no evidence of bribery – as did an International Chamber of Commerce tribunal. In May 1993 Westinghouse and Burns & Roe were cleared of civil bribery charges by a District Court for New Jersey jury. Charges were also brought before the Philippines' Sandiganbayan anti-graft court in 1987 against Herminio Disini, who brokered the Bataan contract, and former President Marcos and his wife Imelda Marcos. In April 2012 the court absolved the Marcoses due to lack of evidence, but ruled that Disini had received $50 million in commissions from Westinghouse and had exerted undue influence in the awarding of the Bataan contract.

In April 2007, the Philippine government made the final payment for the plant. The government was considering converting it into a natural gas-fired power plant, but this seemed impractical, and it has simply been maintained at a cost of some $800,000 per year.

In 2008 an International Atomic Energy Agency (IAEA) mission commissioned by the government advised that Bataan could be refurbished and economically operated for 30 years. Refurbishment, with upgrade of safety and instrumentation & control systems, was estimated to cost $800 million to $1 billion. The IAEA was also to recommend a policy framework for nuclear power development in the country. In December 2008 the National Power Corporation (Napocor) commissioned Korea Electric Power Corp (Kepco, parent company of Korea Hydro & Nuclear Power) to conduct an 18-month feasibility study on commissioning Bataan. One factor in choosing Kepco for this was its experience with Kori 2, a very similar unit in Korea. Its preliminary recommendation in December 2009 was that Bataan should be refurbished. Meanwhile, Toshiba expressed interest in rehabilitating the plant. In May 2013 Napocor urged the government to refurbish and commission the plant to address power shortages. It estimated the $1 billion cost as being one-third of building equivalent coal-fired capacity. Following sharp rises in electricity prices over 2014 to 2016 the DOE was reported to be studying the prospect of reviving the project with South Korean help.

The possible revival of the Bataan plant remains under consideration, but the DOE is also looking into building a new nuclear plant using small modular reactor (SMR) technology.

In November 2022 DOE Undersecretary of Energy Sharon Garin announced that it would cost US$2.3 billion to refurbish and startup Bataan and that the DOE hopes to utilize a portion of its 2023 budget to commence a third-party assessment of the facility. Garin said the country is also weighing the option of building SMRs.

In January 2023 the DOE released an updated priority list for 2023 and stated that it would study the possible inclusion of nuclear power to the power mix by developing SMRs.

In October 2024 it was announced that Korea Hydro & Nuclear Power (KHNP) would complete a technical and economic feasibility study on the potential rehabilitation of the Bataan plant. The costs of the study will be borne by KHNP. The study commenced in January 2025 in two phases: first assessing the plant's current condition; then evaluating refurbishment options.

Small modular reactors

In May 2023 it was announced that NuScale planned to conduct a siting study in the Philippines.

In November 2023 Philippines' largest utility, Manila Electric Company, signed a cooperation agreement with Ultra Safe Nuclear Corporation to study the potential deployment of one or more microreactors.

Fuel cycle

Waste

In 1998 the Philippines signed the Joint Convention on the Safety of Spent Fuel Management and Radioactive Waste Management.

Public opinion

According to the DOE, negative perception of nuclear energy is prevalent. In PEP 2020-2040 it acknowledges the need to improve

public awareness and acceptance to allow for a nuclear energy programme to move forward. However, among respondents in the Philippines to Radiant Energy Group's 2023 PACE index, more support the use of nuclear energy than oppose it.

Research & development

A 1 MW open pool general-purpose PRR-1 research reactor in Quezon City, constructed by General Atomics of the USA, reached criticality in August 1963. In 1984, the Philippine Nuclear Research Institute (PNRI) decided to convert and upgrade the reactor into a 3 MW TRIGA Mark III reactor. It was shut down in 1988, leaving the country with no operating nuclear facility.

In 2014, a proposal was accepted to utilize fuel rods from PRR-1 for training and education. The fuel is a uranium-zirconium hydride alloy manufactured by General Atomics.

The IAEA has been supporting through a series of technical cooperation projects. In the first, which launched in 2016, the IAEA assisted PNRI to build capacity in reactor design, neutron dosimetry and regulatory matters related to research reactors. A second cooperation project followed in 2020 and is ongoing to further build capacity, particularly in reactor engineering and operation, reactor utilization and development of a reactor training programme to sustain local capacity-building activities.

The new Philippine Research Reactor-1 Subcritical Assembly for Training, Education and Research (SATER) facility is housed in the PRR-1 building and will remain in a subcritical state.

On 20 June 2022, PNRI began loading 44 TRIGA nuclear fuel rods into the core of the PRR-1 SATER, signaling the start of the nuclear commissioning of the reactor.

Regulation, safety and non-proliferation

The Philippine Nuclear Research Institute (PNRI), formerly the Philippines Atomic Energy Commission (PAEC), acts as both the nuclear regulatory authority and the entity responsible for research and development of nuclear sciences and technology.

In July 2017 a bill was introduced to Congress to create an independent nuclear safety regulator – the Philippine Nuclear Regulatory Commission (PNRC) – that would decide on issues

affecting public health and safety, protection of the environment, and nuclear security and safeguards. Currently the PNRI carries out promotional and regulatory functions under the Department of Science and Technology (DOST). The fiscal provisions required for the passage of the bill were approved in August 2021. In September 2025 President Marcos signed the Philippine National Nuclear Energy Safety Act (RA 12305), establishing the Philippine Atomic Energy Regulatory and Safety Authority (PhilATOM) as the country's independent nuclear regulator, separating regulatory functions from PNRI's promotional role.

Non-proliferation

The Philippines' safeguards agreement with the IAEA under the Nuclear Non-Proliferation Treaty (NPT) entered into force in 1974 and the Additional Protocol in February 2010.

Notes & references

Republic of the Philippines Department of Energy, Philippine Energy Plan 2020-2040 (October 2021)
The President of the Philippines, Executive Order No. 164 (February 2022)

Related information

Asia's Nuclear Energy Growth